

ICT 1404 Mathematics & Statistics for Computing
Assignment 02

1. Suppose that the time taken by programmers to complete a computer program form a normal distribution with mean $\mu = 50$ hours and standard deviation $\sigma = 10$ hours. Find the probability that the time required to complete the program in hours,
- (a) Between 40 and 55, (b) between 43 and 48, (c) over 60,
(d) Less than 42.

What is the maximum time out of the 15% percent of the programmers who has completed the program in a low time period?

2. Suppose 1 percent of the students on average are left handed. Find the probability that
(a) exactly 2 students (c) 3 or more students are left handed if the number of left handed people are in a Poisson distribution.

- 3 From the following data of hours worked in a factory (x) and output units (y), determine the regression line of y on x, coefficient of determination, R^2 and interpret the goodness of fit of the regression line.

Hours (X)	80	79	83	84	78	60	82	85
Production (Y)	300	302	315	330	300	250	300	340

Submit on or before 25/02/ 2025